

Quantifying ETF Relationships Under Macroeconomic News: Statistical Analysis Using Market Indicators

Team: Honore Alexander, Owen Holt, Pranavi Khanna, Dylan Lim, Yihong Li, Ethan Lee, Dan Firstenberg, Hyeongseung Nam, Oscar Pineda, Kevin Zhang, Zachary Chan, Vicente Aguayo

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Abstract

Within recent years, machine learning has emerged as one of the most powerful tools in predicting stock performances for investors. Such machine learning algorithms integrate market data, financial reports, and macroeconomic data to predict stock movement and volatility; however leading methods systematically overreact to macroeconomic news or fail to take it into account completely. A central aim of our project is to quantify how exchange-traded fund (ETF) relationships reflect underlying macroeconomic conditions, and how these relationships influence individual stock behavior. To address this limitation, we constructed a data management process linking historical company-specific news headlines to corresponding stock performance data across 60 individual tickers.

(Add 1-2 more sentences for method once done with ML side) (Add 2-3 more sentences on results)

Keywords: Machine Learning, Exchange-Traded Funds(ETF's), Batch Gradient Descent, Ordinary Least Squares

1 Introduction

Machine learning has become a staple in the field of quantitative finance, especially for the prediction of stock prices, market trends, and volatility. These models are exceptionally powerful at detecting and representing complex patterns in large datasets, and often outperform traditional analysis techniques (such as econometrics) from other fields such as economics and statistics[13].

One of the biggest challenges is training a model that can accurately predict stock prices, given the instability and unpredictability of the stock market. Obvious approaches, such as using historical stock prices as the training dataset, have been attempted but these endeavours are often difficult due to factors such as dataset noise which makes data preprocessing very difficult. Naturally, stock prices and market prices are also often affected by other factors such as unexpected economic or social events, and investors' psychological behaviors [14].

When it comes to modeling and predicting stocks behaviors based on news events, it's an extremely complex endeavor due to factors such as high variance of macroeconomic event impact, diversity of economic news publications, and complex interactions beyond just macroeconomic news and stock prices. For example, a model based on a Long Short-Term Memory (LSTM) architecture trained in 2023 attempting to predict individual stock prices based on stock-related news only achieved an accuracy of around 55% despite surpassing baseline models [12], still beating, which is only a bit better than a coin flip. This approach can be limiting because individual stock prices can easily fluctuate due to factors such as company-specific news or events.

2 Methods

3 Results

4 Discussion

5 Conclusion

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Author Contributions

- **Honore Alexander**
- **Owen Holt**
- **Pranavi Khanna**
- **Dylan Lim**
- **Yihong Li**: Researched sources, drafted report sections, set up and created frontend interface for model demo
- **Ethan Lee**
- **Dan Firstenberg**
- **Hyeongseung Nam**
- **Oscar Pineda**: Researched sources for the report, contributed to writing the report
- **Kevin Zhang**
- **Zachary Chan**
- **Vicente Aguayo**